$$u = \sum_{j=1}^{d} a_j \vec{v}_j = V\vec{a}$$

$$V = (v_1 \,|\, v_2 \,|\, \ldots \,|\, v_d)$$

$$S = \hat{C}\,(n-1)$$

$$SV = V\Lambda$$

$$V^T S V = \Lambda$$

as V is orthonormal

$$u^t S u = (Va)^t S V a$$

$$= a^t V^T S V a$$

$$= a^t \Lambda a$$

$$= \sum_{j=1}^{d} a_j^2 \lambda_{jj}$$

$$\alpha_{ij} = u_j^t y_i = u_j^t (x_i - \bar{x})$$

$\hookrightarrow$ this is called the j^{th} eigencoefficient of x_i

Eigenfaces

TRAINING PHASE

$x_1 \cdots x_N$ — N different vectorized training images of G diff. people

$l_1 \cdots l_N$ — label for each image giving the identity of the person

$\bar{x}$ = mean vector

Cov. matrix $= \hat{C} = \frac{1}{N-1} \sum_{i=1}^{N} (\vec{x}_i - \bar{x})(\vec{x}_i - \bar{x})^t$ $= ZZ^T$

Find eigenvectors of $\hat{C}$ corresp. to K largest eigenvalues. $\longrightarrow V \quad d \times K$

$\vec{\alpha}_i = V^T(x_i - \bar{x})$ eigencoeff. vector

Output of training phase $= \bar{x}, V, \{\alpha_i\}_{i=1}^{N}$

Testing phase

1. Given a query image q of one of the G different people in the training d/b.
2. Aim: determine identity
3. $\alpha_q = V^T(q - \bar{x}) \longrightarrow \alpha_q \in R^k$
4. Out of all N training image, determine the one for which $\|\alpha_i - \alpha_q\|^2$ is minimum. Then label of $q = l_i$

$d \times 1$ $\quad$ $d \times 1$

$$\tilde{x}_i = \bar{x} + \sum_{l=1}^{k} \vec{V}_l \, \alpha_{il} \qquad k < d$$

$d \times 1$

$$x_i \neq \tilde{x}_i$$

$$x_i = \bar{x} + \sum_{l=1}^{d} \vec{V}_l \, \alpha_{il}$$

$$\|x_i - \tilde{x}_i\|^2 = \left\| \sum_{l=1}^{d} V_l \alpha_{il} - \sum_{l=1}^{k} V_l \alpha_{il} \right\|^2$$

$$= \left\| \sum_{l=k+1}^{d} V_l \alpha_{il} \right\|^2$$